Mursalin Habib

PhD Student · Rutgers University

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Research Interests

I am broadly interested in all things theoretical computer science. My current research is focused on the theory of error-correcting codes.

Education

Rutgers University — *New Brunswick, USA*
PhD in Computer Science — *2023 – present*

Bangladesh University of Engineering and Technology — *Dhaka, Bangladesh*
BSc in Computer Science and Engineering — *2016 – 2021*

Publications

Ulam Rank Aggregation Is Hard to Approximate for Four Rankings
with Sk Ruhul Azgor, Diptarka Chakraborty, Le Van Cuong, Debarati Das and Tien Long Nguyen
Proceedings of the 18th Innovations in Theoretical Computer Science (**ITCS**), 2027.

An Optimal Algorithm for Binary Closest String
with Nick Fischer
Proceedings of the SIAM Symposium on Simplicity in Algorithms (**SOSA**), 2027.

Constant Rate Isometric Embeddings of Hamming Metric into Edit Metric
with Sudatta Bhattacharya, Sanjana Dey, Elazar Goldenberg, Bernhard Haeupler, Karthik C. S. and Michal Koucký
Proceedings of the International Colloquium on Automata, Languages, and Programming (**ICALP**), 2026.

Algorithmic Improvements to List Decoding of Folded Reed–Solomon Codes
with Vikrant Ashvinkumar and Shashank Srivastava
Proceedings of the ACM-SIAM Symposium on Discrete Algorithms (**SODA**), 2026.

Hardness of Median and Center in the Ulam Metric
with Nick Fischer, Elazar Goldenberg and Karthik C. S.
Proceedings of the European Symposium on Algorithms (**ESA**), 2025.

Explicit Good Codes Approaching Distance 1 in Ulam Metric
with Elazar Goldenberg and Karthik C. S.
Proceedings of the IEEE International Symposium on Information Theory (**ISIT**), 2024.
IEEE Transactions on Information Theory, Volume 71, Issue 7, July 2025.

Selected Talks

Algorithmic Improvements to List Decoding of Folded Reed–Solomon Codes

2026 ACM-SIAM Symposium on Discrete Algorithms, Vancouver, BC, Canada — *January 2026*
NYC Student Theory Day, New York, NY, USA — *November 2025*

Hardness of Center and Median in the Ulam Metric

2025 European Symposium on Algorithms, Warsaw, Poland — *September 2025*

Constant Rate Isometric Embeddings of Hamming Metric into Edit Metric

Columbia Theory Lunch, New York, NY, USA — *April 2025*

NYU Theory Seminar, New York, NY, USA *April 2025*
Rutgers/DIMACS Theory of Computing Seminar, Piscataway, NJ, USA *January 2025*

Explicit Good Codes Approaching Distance 1 in Ulam Metric

2024 IEEE International Symposium on Information Theory, Athens, Greece *July 2024*
DIMACS Workshop on Efficient Algorithms for High Dimensional Metrics *May 2024*

Academic Honors & Awards

2026 Rizvi Research Excellence Award, Rutgers Computer Science Department
2026 Research and Travel Award, Rutgers School of Graduate Studies
2024, 2025 Outstanding Teaching Assistant Award, Rutgers Computer Science Department
2024 ISIT 2024 Student Travel Grant, IEEE Information Theory Society
2017–2021 University Merit Scholarship, Bangladesh University of Engineering and Technology
2017–2021 Dean's List Award, Bangladesh University of Engineering and Technology
2015 Champion, Bangladesh Mathematical Olympiad

Service

Served as Referee

Conferences: QIP 2027, SOSA 2027, FSTTCS 2026, RANDOM 2026, MFCS 2026, ESA 2026, ICALP 2026, ITCS 2026, SOSA 2026, CCC 2025, ESA 2024, WALCOM 2023, WALCOM 2021, COCOON 2021, IWOCA 2020.

Journals: Theoretical Computer Science, Algorithmica, Journal of Graph Algorithms and Applications.

Other Services

2026 Co-organizer, Rutgers/DIMACS Theory of Computing Seminar
2022 Deputy Leader, Team Bangladesh, European Girls' Mathematical Olympiad
2021 Observer, Team Bangladesh, International Mathematical Olympiad
2020–2021 Academic Coordinator, Bangladesh Mathematical Olympiad

Research Visits

University of Illinois Urbana-Champaign *Illinois, USA*
Host: Fernando Granha Jeronimo *Summer 2026*

Institute for Computer Science, Artificial Intelligence and Technology (INSAIT) *Sofia, Bulgaria*
Host: Amir Abboud and Bernhard Haeupler *Fall 2024*

Teaching Experience

At Rutgers

Spring 2025 Automata and Formal Languages (01:198:452), Teaching Assistant
Spring 2024 Discrete Structures I (01:198:205), Teaching Assistant
Fall 2023 Automata and Formal Languages (01:198:452), Teaching Assistant

Other Teaching

July 2024 DIMACS Tutorial on Fine-grained Complexity, Teaching Assistant